

Linea Yield Management Audit Summary

Prepared by Cyfrin

Version 2.1

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Linea on Linea Yield Management. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated August 20, 2026, was delivered to Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 10 days, the Cyfrin team conducted an audit on the Linea Yield Management code provided by Linea. The findings consist of 1 High, 7 Medium & 6 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Linea Yield Management
Repository	linea-monorepo
Commit	4360e31fa8c8...
Fix Commit	9c16ce37d2dc...
Audit Timeline	Nov 3rd - Nov 14th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	7	7	0
Low Risk	6	2	4
Informational	9	6	3
Gas Optimizations	2	2	0
Total	25	18	7

6 Audit Scope

This was an incremental audit of the new “Yield Management” functionality, including its integration with the new Lido V3 protocol, and related changes to existing contracts. The audit scope was limited to:

```
contracts/contracts/messageService/l1/L1MessageService.sol
contracts/contracts/lib/YieldManagerPauseManager.sol
contracts/contracts/messageService/lib/MessageHashing.sol
contracts/contracts/yield/libs/ValidatorContainerProofVerifier.sol
contracts/contracts/yield/LidoStVaultYieldProvider.sol
contracts/contracts/yield/LidoStVaultYieldProviderFactory.sol
contracts/contracts/yield/YieldManager.sol
contracts/contracts/yield/YieldManagerStorageLayout.sol
contracts/contracts/yield/YieldProviderBase.sol
contracts/contracts/LineaRollup.sol
contracts/contracts/LineaRollupBase.sol
contracts/contracts/LineaRollupYieldExtension.sol
```

7 Deployment Bytecode Verification

The new functionality, including mitigations from all audits, was merged into the `main` branch at commit 9c16ce3.

At the conclusion of the audit, after all mitigations were concluded, Cyfrin was asked to verify that at commit 25e323d the locally compiled bytecode matched the expected bytecode listed at that commit and the bytecode deployed to Ethereum mainnet. The following reproducible process was used.

First, prepare the repository:

```
gh repo clone Consensys/linea-monorepo
cd linea-monorepo
git checkout 25e323d055dec40ef167a190c71c30aa9bf92c23
pnpm install --ignore-scripts
cd contracts
npx hardhat test
```

Second, diff the locally compiled artifacts against the expected artifacts in `deployments/bytecode/2026-01-14`:

```
diff build/contracts/LineaRollup.sol/LineaRollup.json \
  deployments/bytecode/2026-01-14/LineaRollup.json
diff build/contracts/yield/LidoStVaultYieldProviderFactory.sol/LidoStVaultYieldProviderFactory.json \
  deployments/bytecode/2026-01-14/LidoStVaultYieldProviderFactory.json
diff build/contracts/yield/libs/ValidatorContainerProofVerifier.sol/\
ValidatorContainerProofVerifier.json \
  deployments/bytecode/2026-01-14/ValidatorContainerProofVerifier.json
```

Finally, for each deployed contract, fetch the on-chain bytecode with `eth_getCode` and compare it with the `deployedBytecode` of the local build artifact, after stripping the CBOR metadata and ignoring the positions where the local bytecode holds zero bytes for immutable values. The comparison succeeded for all five contracts:

- `LineaRollup`: 0x04728BF704a716C26F9EF4085013b760AC885631
- `YieldManager`: 0x751236A1aFC11B7F1A7630fe87b0Bd96AC5203C4
- `LidoStVaultYieldProvider`: 0x486D8cADc10489B30b64c890aEc747F1220eEEC3
- `LidoStVaultYieldProviderFactory`: 0xE4FC9F1A8cB97fEAd3C2b37c11AD5B1C2eF73959
- `ValidatorContainerProofVerifier`: 0x2309C45d44105928b483f608dD6140Fb65f3EBde

No immutable differences were reported for `LineaRollup`. For the other four contracts, the only differences were at the positions of known immutable values: the `GI_STATE_ROOT` constant in `ValidatorContainerProofVerifier` (deployed value `0b03` where the local bytecode holds `0000`, at byte ranges 427-428, 1544-1545 and 1724-1725), and the addresses of the following contracts:

```
LineaRollup                    0xd19d4b5d358258f05d7b411e21a1460d11b0876f
Staking Vault Factory          0x02ca7772ff14a9f6c1a08af385aa96bb1b34175a
Vault Hub                      0x1d201be093d847f6446530efb0e8fb426d176709
TransparentUpgradeableProxy    0xeb63cabdd78537b9b72a2afb573f7caa91bd8d94
ValidatorContainerProofVerifier 0x2309c45d44105928b483f608dd6140fb65f3ebde
Lido and stETH token           0xae7ab96520de3a18e5e111b5eaab095312d7fe84
```

LineaRollup update (PR3654). After the original yield functionality was deployed, Cyfrin reviewed PR3654, a `LineaRollup` update in the repository now hosted at `LFDT-Lineth/lineth-monorepo`. The PR was merged at commit 82af194, tagged `contract-audit-2026-07-30`, and the resulting `LineaRollup` implementation was deployed on Ethereum mainnet at 0x052b73d934E9412045Bf731574463Fd026D74645.

At commit 82af194, Cyfrin verified that the locally compiled `LineaRollup` artifact matched both the expected artifact committed to the repository and the deployed runtime bytecode, using the following process:

```
gh repo clone LFDT-Lineth/lineth-monorepo
cd lineth-monorepo
git checkout 82af194f8c680abd238bdd5f7c85acabf073bdbb
nvm install
nvm use
corepack enable
pnpm install --ignore-scripts
(cd node_modules/.pnpm/c-kzg@4.1.0/node_modules/c-kzg && npx node-gyp rebuild)
pnpm -F contracts run build
diff contracts/build/src/rollup/LineaRollup.sol/LineaRollup.json \
  contracts/deployments/bytecode/2026-07-29/LineaRollup.json
```

The local and expected artifacts are byte-identical and have the same SHA-256 hash:

```
51049d4c76fd8e0fb7378ae0173c6af057e949edbae4679f9377844df946c865
```

The unit and integration test files modified by PR3654 were then run from the `contracts` directory:

```
cd contracts
pnpm exec hardhat test test/hardhat/yield/unit/LineaRollupYieldExtension.ts \
  test/hardhat/yield/integration/YieldManager.integration.ts
```

The tests completed with 49 passing and 1 pending.

The on-chain runtime bytecode was fetched at Ethereum block `25795292` and compared with the expected artifact:

```
cast code 0x052b73d934E9412045Bf731574463Fd026D74645 --rpc-url <mainnet-rpc> > LineaRollup.mainnet.txt
diff <(jq -r .deployedBytecode deployments/bytecode/2026-07-29/LineaRollup.json) LineaRollup.mainnet.txt
```

The on-chain runtime bytecode is byte-identical to the expected `deployedBytecode`, including the trailing metadata (CBOR) hash. Its Keccak-256 hash is:

```
0x5a81c1fedcf9fa79a757c3871bd636f27accb5a963f564e0e8908629e77dcb52
```

Compared with the preceding `deployments/bytecode/2026-02-20/LineaRollup.json` artifact, the only executable runtime bytecode difference is a single opcode at offset `0x1d15`, matching the source change in PR3654; the remaining difference is the expected metadata hash change. The verified artifact was compiled with Solidity `0.8.33`, EVM version `osaka` and optimizer runs set to `200` for `LineaRollup`.

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

Linea is a Type 2 zkEVM L2 rollup with full EVM equivalence which aims to provide the “Ethereum experience” at scale and has been previously audited by Cyfrin.

10 Attestation

Cyfrin attests that it performed a security review of the Linea Yield Management code provided by Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin’s repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-08-20-cyfrin-linea-yield-manager-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin’s full audit report with all findings provided to Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=316b1963945933d3e2b8beda3a0ed2e46ff691e2353f39863554d49d665e60d3
F=2026-08-20-cyfrin-linea-yield-manager-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.